Marketplace Leak Audit: Memo

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QUESTION

Where does an online marketplace lose money and customer trust? I checked 99,441 orders for late delivery and for payments that do not match what was ordered.

DEFINITIONS

- **Late:** a delivered order that arrived after the promised date. Late % = late orders / delivered orders.
- **Mismatch:** amount paid differs from item price plus shipping by more than about 1.5 cents. Small gap is up to 1 BRL, large gap is over 1 BRL.
- **Missing items:** an order with a payment but no item rows. Counted separately, not as a mismatch.

FINDINGS

Finding	Number	So what
1. Late orders get low reviews	8.11% of delivered orders were late. Average review 2.57 late vs 4.29 on time.	Lateness is where customer trust breaks. Late and low scores go together; this does not prove lateness caused them.
2. Lateness is concentrated	Late share: AL 23.93%, MA 19.67%, PI 15.97%. SP 5.89%.	The delivery promise looks unrealistic in a few northern and northeastern states, not everywhere.
3. Payment mismatches are small	303 orders (0.3%), 3,269.22 BRL in total gaps (249 large, 54 small).	Little money lost to wrong payments. Not the main leak.
4. Payments with no items	775 orders, 162,591.95 BRL paid with no item records. About 98% of the money in non-matching orders.	The biggest gap is a data problem, not a payment error. Cause not investigated.

RECOMMENDATION

- **Logistics team:** set longer promised delivery dates for AL, MA and PI and review carrier performance there. Side effect: longer promises may lower conversion.
- **Data team:** find out why 775 orders have payments but no item records, starting with their order statuses (canceled, created, invoiced, shipped, unavailable).
- **Next test:** compare actual and promised delivery days by state to see where the promise is least realistic.

LIMITS

- Public data from 2016 to 2018. It does not describe any company today.
- Association only: late orders score lower, but I did not test cause.
- Vouchers and installments may explain some payment gaps. I did not test this.
- Small states (for example RR with 46 orders) swing a lot on a few orders.

CHECK

Excel and Power BI agree on Orders (99,441), Late % (8.11%) and Mismatch Amount (3,269.22 BRL).

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